

Optimal-transport flow matching as an alternative to CycleGAN

Unpaired PYTHIA → JEWEL translation of clean calorimeter jet images

October 2026 (CycleGAN at 2 and 8 hours; 24 hours pending)

seed 0; 20k held-out jets per sample; full log: FLOW_NOTES.md, results: docs/flow/translation

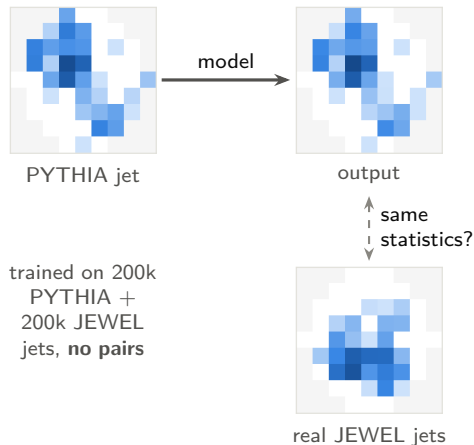
The question

Can optimal-transport flow matching replace a CycleGAN for **unpaired** jet translation: clean PYTHIA (vacuum) $\rightarrow$ JEWEL-like (medium)?

Two things must hold:

1. the outputs look like real JEWEL jets, statistically;
2. each output still depends on its own input jet.

There is no true JEWEL partner for a PYTHIA jet: any unpaired map is one of many with the same distributions. So this tests distributions and dependence, not a per-jet medium modification.



Two ways to learn an unpaired map

— OT-FM [1, 2]

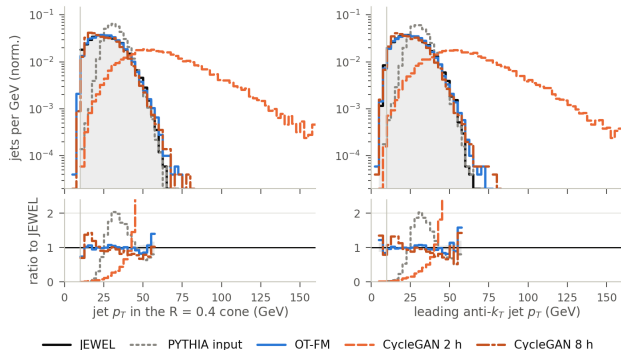
- pair each batch of 256 PYTHIA and 256 JEWEL jets by optimal transport (least change in log energy)
- regress the velocity along straight lines between the pairs: a plain mean-squared-error loss, no adversary
- translate: solve the flow ODE, 128 network calls

— **CycleGAN** [3–5]: the repository's UVCGAN2, which UVCGAN-S builds on

- two generators (PYTHIA $\rightarrow$ JEWEL, JEWEL $\rightarrow$ PYTHIA) and two discriminators
- adversarial, cycle-consistency and identity losses
- translate: one pass of the PYTHIA $\rightarrow$ JEWEL generator

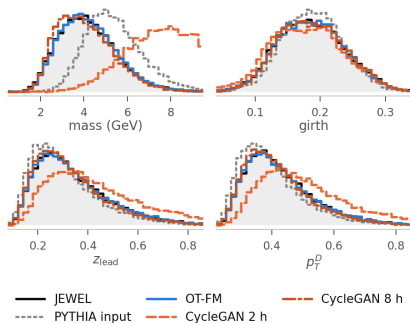
Held equal: the same 200k + 200k training jets (the leading $R = 0.4$ jet, 53 towers on a 16×16 grid), the same log-energy coordinates, the same generator network (the UVCGAN-S ViT-ModNet, 32M parameters), seed 0, and **2 hours on one A6000**. CycleGAN also continues to 8 and 24 hours.

Jet p_T spectra



- **OT-FM** reproduces the JEWEL jet p_T spectrum within statistics (cone and re-found jets), although its inputs start near 20 GeV
- **CycleGAN** at the same 2 hours: p_T about doubled (mean 65 GeV; JEWEL 27); at 8 hours on the JEWEL scale (mean 26.0 GeV), but 5 times **OT-FM**'s distance (W_1/σ 0.12 against 0.025); its energy scale swung for 5 hours before settling

Jet substructure

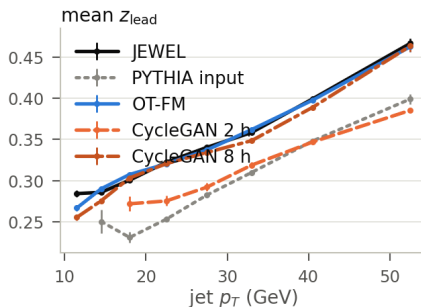
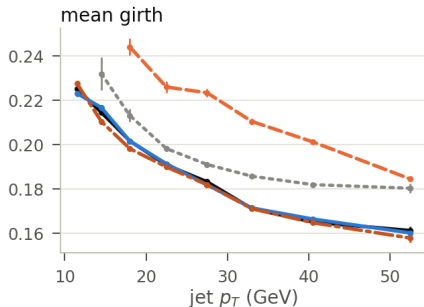


	JEWEL vs JEWEL	PYTHIA input	random JEWEL jet	OT-FM	CycleGAN 2 h	CycleGAN 8 h
p_T	0.011	0.55	0.019	0.025	3.91	0.12
mass	0.014	0.88	0.024	0.021	3.33	0.11
girth	0.009	0.10	0.011	0.012	0.14	0.015
z_{lead}	0.019	0.23	0.011	0.008	0.58	0.076
p_T^D	0.018	0.26	0.010	0.010	0.66	0.073
z_g	0.012	0.034	0.005	0.017	0.12	0.017
R_g	0.008	0.16	0.010	0.012	0.22	0.011

Distance to 20k held-out JEWEL jets ($W1 / \sigma$, lower is better; jets with $p_T \geq 10$ GeV). Bold: as close as a second JEWEL sample, within 3 standard deviations.

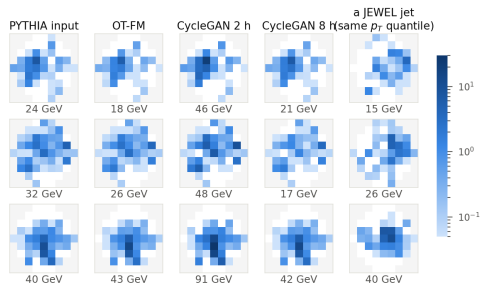
- **OT-FM**: all seven observables as close to JEWEL as a second JEWEL sample
- **CycleGAN** at 2 hours: no observable closer to JEWEL than the input (cores far too hard). At 8 hours 6 of 7 move toward JEWEL; girth, z_g , R_g as close as a second JEWEL sample; p_T , mass, p_T^D , z_{lead} 5-9 times **OT-FM's** distance

Jet shapes at fixed p_T



- at the same p_T , JEWEL jets are **narrower** and **harder** (higher z_{lead}) than PYTHIA jets
- **OT-FM** follows both trends in every p_T bin
- **CycleGAN** at 2 hours: girth at fixed p_T further from JEWEL than the input's; at 8 hours both trends, less closely than **OT-FM** (W_1/σ at fixed p_T 0.04 and 0.06 against 0.02)

Does each output keep its input jet?

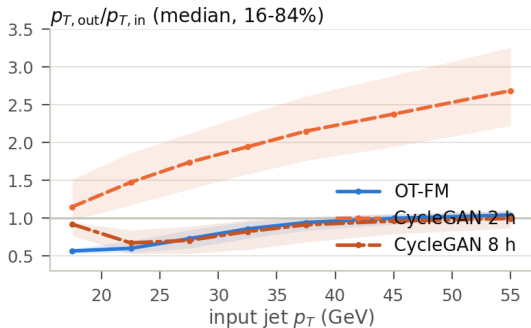


	random JEWEL jet	OT-FM	CycleGAN 2 h	CycleGAN 8 h
p_T	0.01	0.94	0.85	0.82
girth	0.00	0.99	0.99	0.99
mass	0.01	0.91	0.85	0.75
core fraction	0.00	0.99	0.99	0.99
leading tower	0.00	0.92	0.92	0.87
own input preferred	50%	100%	100%	100%

Input-output correlation over 20k test jets; own input: share of outputs closer in shape to their own input than to another, random input jet.

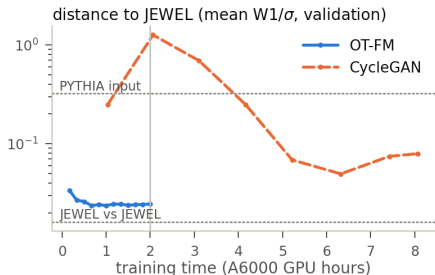
- **OT-FM**: correlations 0.91-0.99; every output is closer to its own input than to another jet
- **CycleGAN** keeps the input too: correlations 0.75-0.99 at 8 hours, every output closest to its own input

Caveat: the p_T change follows the two samples



- the PYTHIA sample starts near 20 GeV (presumably its generator-level jet requirement); JEWEL reaches down to 10 GeV
- **OT-FM** lowers 20-25 GeV jets by 40% and leaves jets above 40 GeV unchanged: a map between two differently selected samples, not a quenching mechanism
- **CycleGAN** at 2 hours raised p_T by 1.2-2.6; at 8 hours it makes nearly the same spectrum map as **OT-FM** (median output 16 GeV for 20-25 GeV inputs, 41 GeV for 40-50 GeV)

Training cost



	GPU h	updates	updates/s	network calls	ms / jet
OT-FM	2.0	123k	17.0	128	6.58
CycleGAN 2 h	2.1	20k	2.7	1	0.05
CycleGAN 8 h	8.1	76k	2.6	1	0.05

One A6000; training time without checkpoints; latency at batch 2000.

- **OT-FM**: 123k updates in 2 GPU hours (17 per second), one network pass each
- **CycleGAN**: 2.6-2.7 updates per second (six generator passes and two discriminators per update): 20k updates in 2 hours, 76k in 8; one network pass per jet, 130 times faster to apply

Summary

- **At the same 2 GPU hours: OT-FM** (no pairs) produces jets statistically indistinguishable from held-out JEWEL (p_T , 7 substructure observables, their p_T dependence, re-found jets), and each output stays tied to its input
- **CycleGAN** (the repository's UVCAN2, same data and generator network) does not at 2 hours: p_T off by a factor 2. At 8 hours ($4\times$ the budget) it is useful by the same rule: 6 of 7 observables toward JEWEL, girth and soft drop as close as a second JEWEL sample, inputs kept; p_T , mass, p_T^D and z_{lead} still 5-9 times further from JEWEL than **OT-FM's**
- its training swung for 5 hours (the 1-hour checkpoint erred low, 2 hours high); 24 hours pending
- **OT-FM's** p_T change reflects how the two samples were selected, not quenching; no per-jet truth exists

References

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